Time Complexity

Qn 1)Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n
Output:
Print the value of the counter variable

For example:

Input	Result
9	12

Ans:

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int count=0;
    int i=1;
    count++;
    int s=1;
    count++;
    while(count++,s<=n){
        i++;
        count++;
        s+=i;
        count++;
    }
    printf("%d",count);
    return 0;
}
```

Testcase:

Input	Expected	Got
9	12	12
4	9	9

Testcase passed successfully!

Qn 2)Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
```

```
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.
Input:
A positive Integer n
Output:
Print the value of the counter variable

Ans:

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int count=0;
     for(int i=1;count++, i<=n;count++, i++){
       for(int j=1;count++, j<=n; j++){
           count++;
           break;
       }
       count++;
     }
     count++;
     printf("%d",count);
     return 0;
 }
```

Testcase:

Input	Expected	Got
2	12	12
1000	5002	5002
143	717	717

Testcase passed successfully!

Qn 3)Convert the following algorithm into a program and find its time complexity using counter method.

```
 Factor(num) {
 {
    for (i = 1; i <= num;++i)
    {
     if (num % i== 0)
        {
```

```
          printf("%d ", i);
        }
     }
  }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:
A positive Integer n
Output:
Print the value of the counter variable

Ans:

```
#include<stdio.h>
int main(){
    int num;
    scanf("%d",&num);
    int i;
    int count=0;
    for(i=1;count++,i<=num;++i){
        if(num%i==0){
            count++;
        }
        count++;
    }
    printf("%d",count);
    return 0;
}
```

Testcase:

Input	Expected	Got
12	31	31
25	54	54
4	12	12

Testcase passed successfully!

Qn 4)Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n
Output:
Print the value of the counter variable

Ans:

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int c=0;
    for(int i=n/2;c++,i<n;c++,i++){
        for(int j=1;j<n;c++,j=2*j){
            for(int k=1;c++,k<n;c++,k=k*2){
            }
        }
    }
    c++;
    printf("%d",c);
    return 0;
}
```

Testcase:

Input	Expected	Got
4	30	30
10	212	212

Testcase passed successfully!

Qn 5)Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n
Output:
Print the value of the counter variable

Ans:

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int rev=0,remainder;
    int count=0;
    while(count++,n!=0){
        remainder=n%10;
        count++;
        rev=rev*10+remainder;
```

```
        count++;
        n/=10;
        count++;
    }
    count++;
    count++;
    printf("%d",count);
    return 0;
}
```

Testcase:

Input	Expected	Got
12	11	11
1234	19	19

Testcase passed successfully!